SUPPLEMENTARY METHODS

Subjects

Cohort1: We used resting-state EEG data collected by Bahar et al[1] and Ozbek et al[2]. The dataset contains a cohort of 119 participants with 59 individuals diagnosed with early-stage Alzheimer's Disease (AD) (37 female, 21 male), and 60 healthy elderly subjects (36 female, 24 male). Subjects were enrolled in Istanbul Medipol University and/or Dokuz Eylül University. The inclusion criteria for the AD group were based on the National Institute of Neurological and Communicative Diseases and Stroke-Alzheimer's Disease and Related Disorders Association (NINCDS-ADRDA) criteria[3][3], and the Diagnostic and Statistical Manual of Mental Disorders–4th Edition (DSM-IV[4]; American Psychiatric Association, 2000).

Cohort2: This dataset is an open accessible resting state eye closed EEG dataset from the second Department of Neurology of AHEPA General Hospital of Thessaloniki[5]. We used all 36 individuals diagnosed with AD and 29 healthy elderly individuals. The mean age was 66.4 (SD = 7.9) for the AD group and 67.9 (SD = 5.4) for the control group. The dataset is age-matched ($p = 0.3$)[5].

Cohort3: This cohort comprises 85 individuals diagnosed with Mild Cognitive Impairment (MCI) and 30 Healthy Controls (HC)[6]. For our analysis, we age-matched the subjects (N = 30 per group). MCI subjects were recruited from the outpatient memory clinic at the Department of Neurology, Dokuz Eylul University Hospital, while healthy controls were gathered from various community sources with inclusion criteria strictly adhering to no history of neurological/psychiatric abnormalities, no reported history of substance misuse (including drugs and alcohol), and a Mini-Mental State Examination (MMSE) score of ≥ 27[2] . The diagnosis of MCI followed stringent adherence to the National Institute on Aging and Alzheimer's Association (NIA-AA) criteria[3,7].

EEG Data Acquisition

Cohort1 & Cohort3*:* As described previously[1], the resting-state EEG data was acquired in accordance with the international 10-20 electrode placement system, utilizing an array of 30 electrodes placed on an elastic cap. These electrodes were positioned at specific locations, including Fp1, Fp2, F7, F3, Fz, F4, F8, Ft7, Fc3, Fcz, Fc4, Ft8, Cz, C3, C4, T7, T8, Tp7, Cp3, Cpz, Cp4, Tp8, P3, Pz, P4, P7, P8, O1, Oz, and O2. Biauricular reference electrodes (A1 and A2) were affixed to maintain a consistent reference point. Additionally, two electro-oculograms were positioned along the medial upper and lateral orbital rim of the left eye to monitor artifacts such as eye-blinks and eye movements. The data acquisition was conducted at a sampling frequency of 500 Hz. The EEG was recorded for a total of 8 minutes with 4 minutes of eye-open followed by 4 minutes of eye-closed.

Cohort2: As described previously[5], the data acquisition adhered to an international 10-20 electrode placement system with 19 electrodes, including Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, and O2. Additionally, 2 reference electrodes (A1 and A2) were placed on mastoids for impedance check. The recording montages comprised both anterior-posterior bipolar and referential montages using Cz as the common reference. The data acquisition was conducted at a sampling frequency of 500 Hz.

Mini-Mental State Examination (MMSE).

MMSE scores were collected from all three cohorts. Clinical dementia rating (CDR) was also collected for Cohort1 subjects.

Cohort1 & Cohort3: The average MMSE scores were 19.69 (standard deviation, SD = 4.22) for AD, 25.46 (SD = 3.24) for MCI group and 28.42 (SD = 1.65) for healthy elderly subjects. The average MMSE of the AD group was 17.75 (SD = 4.5) in Cohort2.

Data Pre-processing & artifact removal

The raw EEG data was preprocessed in MATLAB using two independent approaches.

Method 1. Detrended & Hampel filtered(det-Hamp): First, the EEG data had presence of power line interference at 50 Hz in all the subjects. To mitigate this recurring interference the raw EEG data was filtered in the range spanning 0.5 – 48 Hz using bandpass filter in MATLAB. Subsequent manual inspection of EEG data showed some baseline wandering which was rectified by detrending (as described below) the EEG data. To eliminate artifacts Hampel filter (described below) was employed. The Hampel filter is particularly useful for identifying and rectifying potential outliers in the data, that might correspond to artifacts in EEG signals. We used one minute eye-closed EEG data for further analysis.

Detrending: The EEG signals were first detrended using the locdetrend.m function in Chronux 2.00 toolbox[8]. The detrending algorithm is based on local linear regression. The algorithm operates locally on moving windows with user-defined length, and step-size, and covers the entire signal through a raster scan. The best fitting line, in a least-squares sense, for each moving window is subtracted from the samples to remove the trend. The duration of the moving window and the step are selected respectively as 0.06 sec and 0.015 sec, respectively.

Hampel Filter: We used the Hampel filter to suppress outlier amplitude values introduced by large transients[9–11]. The formulation of the Hampel filter is based on three-sigma rule of thumb (or 3σ

rule) in statistics. The 3σ rule expresses a conventional heuristic that nearly all values in a population lie within three standard deviations of the mean. In the Hampel filter samples in a discrete signal whose absolute difference from the local median of neighborhood samples are greater than a threshold is replaced by the local median. As described and used previously[9], the filtered signal,

$$y(n) = \{M(n) \; |x(n) - \; M(n)| \; > T \; x(n) \; Otherwise \tag{1}$$

In (1), $M(n)$ is the is the local median of the neighborhood samples $x(n + q)$, $q = \{- Q, ..., 0, ..., + Q\}$ of an arbitrary contextual sample $x(n)$ in the signal $X = \{x(1), x(2), ..., x(n), ...x(N)\}$ with N number of samples in it. The local median $M(n)$ is the median of a window of samples with a length of Q. The window includes $2Q + 1$ samples that are Q samples left to the contextual sample $x(n)$, Q samples right to the contextual sample, and the contextual sample itself. The local median $M(n)$ is,

$$M(n) = \; median\{x(n - Q), ..., x(n), ..., x(n + Q)\} \tag{2}$$

The threshold T in (1) is the product of the standard deviation of the neighborhood samples $\sigma(n)$ and a user-defined parameter β termed number of standard deviations such that $T = \beta\sigma(n)$. The standard deviation of the neighborhood samples $\sigma(n)$ is computed from the median absolute deviation (MAD) of the samples as,

$$\Sigma(n) = 1.4826MAD(n)$$

$$= \; median\{|x(n - Q) - M(n)|, ..., |x(n) - \; M(n)|, ..., |x(n + Q) - M(n)|\} \tag{3}$$

In (3), the MAD, $MAD(n)$ is the median of absolute differences of neighborhood samples $x(n + q)$, $q = \{- Q, ..., 0, ..., + Q\}$ from the local median $M(n)$. In the current implementation of the Hampel filter, number of standard deviations β is selected as 3 (motivated by 3σ rule) and length of the window Q is selected as 1000 samples. Near the endpoints of the signal where $n \leq Q$ and $n \geq N - Q$, the window of neighborhood samples is truncated accommodating only the possible samples.

Method 2. Artifact subspace reconstruction (ASR): The raw EEG data were also pre-processed with the ASR[12] method in MATLAB using the EEGLAB plug-in function' clean_rawdata'. The ASR method consists of three steps: (1) identifying artifact segments and rejecting them based on threshold parameters, (2) finding the artifact-free reference data, and (3) reconstruction of the artifact segments.

1. Identifying artifact segments and rejecting based on threshold parameters: ASR identifies the artifacts based on different criteria, including flatline, burst, channel correlation, line noise. For the

proposed study, the flatline criterion was 5 seconds, the burst criterion was 4 standard deviation (SD), the channel correlation was less than 0.8, and the line noise criterion was 4 SD. The artifact segments may also be detected using principal component analysis (PCA) with an option of spectral weighting using infinite impulse response (IIR) filter to make ASR method blind to desired signals and to increase the sensitivity to artifacts related to muscle.

2. Finding artifact-free reference data: ASR selects clean segments of EEG data based on the signal variance distribution estimated with root-mean-square (RMS) values on a short sliding window of duration 1 second. The RMS values are then z-scored across all windows for each channel. The clean segments are selected within the z-scored range of -3.5 and 5.5 after fitting the distribution into a truncated Gaussian distribution. ASR uses robust estimators to estimate the covariance matrix, mixing matrix, and direction-dependent threshold matrix from the artifact-free data.

3. Reconstruction of the artifact segments: ASR reconstructs the artifact segments with the robust mixing matrix estimated from the clean data.

Feature Extraction and analysis

EEG Features

Forty-one distinct EEG features are evaluated, as described in detail below. These features belong to various domains, including temporal attributes, spectral descriptors, complexity metrics, wavelet characteristics, power ratios, and entropy measures.

Descriptive Statistics

Sample Skewness

The skewness measures the asymmetry of the data around the sample mean. If skewness is negative, the data spreads out more to the left of the mean than to the right. If skewness is positive, the data spreads out more to the right. The skewness of the normal distribution (or any perfectly symmetric distribution) is zero. The skewness s of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$\frac{s=\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})^3}{\left(\sqrt{\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})^2}\right)^3} \quad (1)$$

In (1), $\bar{x}$ is the mean amplitude. We used the term' sample skewness' to distinguish it from spectral skewness[13].

Sample Kurtosis

Kurtosis measures how outlier-prone a distribution is. The kurtosis of the normal distribution is 3. Distributions that are more outlier-prone than the normal distribution have kurtosis greater than 3, whereas distributions that are less outlier-prone have kurtosis less than 3. Some definitions of kurtosis subtract 3 from the computed value, so that the normal distribution has kurtosis of 0. The Kurtosis k of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$\frac{k=\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})^4}{\left(\frac{1}{N}\sum_{n=1}^{N}(x(n)-\bar{x})^2\right)^2} \quad (1)$$

In (1), $\bar{x}$ is the mean amplitude. We used the term' sample kurtosis' to distinguish it from spectral kurtosis[13].

Variance

The Variance σ^2 of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$\sigma^2 = \frac{1}{N-1}\sum_{n=1}^{N}\left(x(n) - \mu_X\right)^2 \quad (1)$$

In (1), μ_X is the mean amplitude[13].

Impulsive Metrics

Impulse Factor

Impulse factor (IF) is the ratio of the largest peak amplitude in the signal and mean amplitude[14]. Impulse Factor $IF(X)$ of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$IF(X) = \frac{x_{max}}{\mu_X},\ given\ x_{max} = x(n) if\ x(n) > x(i),\ \forall\ 1 \leq i \leq N\ and\ n \neq i \quad (1)$$

In (1), x_{max} is the largest peak amplitude in the signal and μ_X is the mean amplitude.

Crest Factor

Crest factor (CF) is the ratio of the largest peak amplitude in the signal and the root mean square (RMS) amplitude[14]. The Crest Factor $CF(X)$ of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$CF(X) = \frac{x_{max}}{X_{RMS}},\ given\ x_{max} = x(n) if\ x(n) > \ x(i),\ \forall\ 1 \leq i \leq N\ \&\ n \neq i\ \ and\ \ X_{RMS} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}[x(n)]^2} \quad (1)$$

In (1), x_{max} is the largest peak amplitude in the signal and X_{RMS} is the RMS amplitude.

Clearance Factor

Clearance Factor (CLF) is the ratio of the largest peak amplitude in the signal and square of the mean value of the square roots of the absolute amplitudes[14]. The Clearance Factor $CLF(X)$ of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$CLF(X) = \frac{x_{max}}{\left(\frac{1}{N}\sum_{n=1}^{N}\sqrt{|x(n)|}\right)^2} \quad (1)$$

In (1), x_{max} is the largest peak amplitude in the signal.

Shape factor

Shape factor (SF) is the ratio of the RMS amplitude and mean of the absolute amplitude values of the signal[14]. The Shape factor $SF(X)$ of a signal $X = \{x(n), x(n), ..., x(n), ...x(N)\}$ comprising N samples in it is,

$$\frac{SF(X) = \sqrt{\frac{1}{N}\sum_{n=1}^{N}[x(n)]^2}}{\frac{1}{N}\sum_{n=1}^{N}|x(n)|} \quad (1)$$

Willison's Amplitude (WAMP)

The Willison's Amplitude (WAMP) is the total number of sample points in the signal where the absolute derivative is above a user-defined threshold. As described previously[15], WAMP can be computed as,

$$WAMP(X) = \sum_{n=1}^{N-1} f(|x(n+1) - x(n)|)\ given\ X = \{x(1), x(2), ..., x(n), ..., x(N)\} \quad (1)$$

In (1), the variable N is the total number of samples in the signal. The function/operator $f(.)$ is[15],

$$f(a) = \{1 \; if \; a > T \; 0 \; Otherwise \tag{2}$$

In (2), the variable T is a user-defined threshold and a is a dummy variable.

Zero-Crossing Rate (ZCR)

Zero-Crossing Rate (ZCR) examines how fast are the amplitude transitions in a signal[16,17]. Consider a signal X with N samples in it.

$$X = \{x(1), x(2), ..., x(n), ..., x(N)\} \tag{1}$$

In (1), $x(n),\ n = 1, 2, ...N$ denotes an arbitrary sample. The ZCR of the signal is[16,17],

$$ZCR(X) = \frac{1}{2N} \sum_{n=2}^{N} |sgn(X(n) - \ sgn(X(n - 1)| \tag{2}$$

The sign function in (2) is,

$$sgn(X(n)) = \{1 \; X(n) \geq 0 \ - 1 \; X(n) < 0 \tag{3}$$

Spectral Descriptors

Alpha Dominant Frequency

Alpha dominant frequency refers to the prominent frequency component in the alpha band (9-12 Hz) of an EEG signal. It is the frequency at which the absolute of the spectral (Discrete Fourier Transform (DFT)) coefficient is higher than those at other frequency bins within the alpha band. In the current implementation[18], the alpha dominant frequency is computed from the half-length spectrum. The spectrum (DFT) Y of a signal $X = \{x(1),\ x(2), ..., x(n), ...x(N)\}$ uniformly sampled at an interval $\frac{1}{f_s}$ and comprising N samples is,

$$Y(k) = \sum_{n=1}^{N} x(n) e^{\frac{-j2\pi kn}{N}},\ k = 1, 2, ..., N \tag{1}$$

The first half of the spectrum $Y(k),\ k = 1, 2, ..., \frac{[(N}{2)+1]}$ corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the component frequencies in a signal, since the second half is just a replication of the first half. The variable f_s represents the sampling frequency. An arbitrary frequency $f(k),\ k = 0, 1, 2, ..., \frac{[(N}{2)+1]}$ can be the dominant frequency if the magnitude of the DFT coefficient $Y(k)$ at $f(k)$ is higher than those at other frequency bins in the preferred alpha band such that,

$$f_{Dom}^{\alpha} = f(k)\ if\ |Y(k)| > |Y(j)|\ given\ f(k) = k\frac{f_s}{N},\ \forall\ \frac{9N}{f_s} \leq k\ \&\ j \leq \frac{13N}{f_s},\ k \neq j,\ and\ 9 \leq f(k) \leq 12 \quad (2)$$

Gamma Dominant Frequency

Like the alpha dominant frequency, gamma dominant frequency f_{Dom}^{γ} refers to the prominent frequency component in the gamma band (31 – 48 Hz) of an EEG signal. The gamma dominant frequency is,

$$f_{Dom}^{\gamma} = f(k)\ if\ |Y(k)| > |Y(j)|\ given\ f(k) = k\frac{f_s}{N},\ \forall\ \frac{31N}{f_s} \leq k\ \&\ j \leq \frac{48N}{f_s},\ k \neq j,\ and\ 31 \leq f(k) \leq 48 \quad (2)$$

Spectral roll-off

The spectral roll-off is the frequency at which the power spectrum crosses 95% of the total power and thus, it provides information regarding the frequency band in which a major part of the signal power is concentrated. The spectral roll-off $f_{Roll-off}$ of a signal $X = \{x(1),\ x(2), \ldots\ldots, x(n), \ldots\ldots x(N)\}$ uniformly sampled at an interval $\frac{1}{f_s}$ and comprising N samples is the arbitrary frequency value $f(i),\ i = 0, 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ that satisfies the criterion[19],

$$f_{Roll-off} = f(i),\ i = 0, 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}\ \ given \sum_{k=1}^{i} P_m(k) = \frac{95}{100} \sum_{k=1}^{\frac{[(N}{2)+1]}} P_m(k) \quad (1)$$

In (1), P_m is a simplified representation of the half-length power spectrum computed from the Discrete Fourier Transform (DFT) of the signal. The spectrum (DFT) Y of the signal X is,

$$Y(k) = \sum_{n=1}^{N} x(n) e^{\frac{-j2\pi kn}{N}},\ k = 1, 2, \ldots\ldots, N \quad (2)$$

The first half of the spectrum $Y(k),\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the component frequencies in a composite, since the second half is just a replication of the first half. The variable f_s represents the sampling frequency. A half-length representation of the PSD estimate can be computed from the half-length spectrum as,

$$P(k) = \frac{|Y(k)|^2}{f_s N},\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]} \quad (3)$$

As Nyquist frequency and DC do not appear twice in the spectrum, the power spectrum is modified to conserve the total power such that the modified power spectrum P_m is,

$$P_m(k) = \{p(k)\ 2p(k)\ ,\ if\ k = 1\ or\ \ k = \frac{[(N}{2)+1]}\ ,\ Otherwise \tag{4}$$

Spectral Centroid

To characterize frequency-dependent changes in the spectrum the spectral centroid, the weighted sum of frequencies, was used. The spectral centroid is also called center of spectral mass. The weight corresponding to any arbitrary frequency depends on the relative amplitude of the spectrum at that frequency. The spectral centroid is the frequency around which most of the signal power (the center of mass or center of gravity of the spectrum) is concentrated[16,17].

$$f_{Centroid} = \frac{\sum_{k=1}^{\frac{[(N}{2)+1]}} f(k)P_m(k)}{\sum_{k=1}^{\frac{[(N}{2)+1]}} P_m(k)} \tag{1}$$

In (1), P_m is a simplified representation of the half-length power spectrum computed from the Discrete Fourier Transform (DFT) of the signal. The spectrum (DFT) Y of the signal X is,

$$Y(k) = \sum_{n=1}^{N} x(n)e^{\frac{-j2\pi kn}{N}}\ ,\ k = 1, 2, \ldots\ldots, N \tag{2}$$

The first half of the spectrum $Y(k),\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the component frequencies in a composite, since the second half is just a replication of the first half. The variable f_s represents the sampling frequency. A half-length representation of the PSD estimate can be computed from the half-length spectrum as,

$$P(k) = \frac{|Y(k)|^2}{f_s N},\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]} \tag{3}$$

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$$P_m(k) = \{p(k)\ 2p(k)\ ,\ if\ k = 1\ or\ \ k = \frac{[(N}{2)+1]}\ ,\ Otherwise \tag{4}$$

Spectral Flux

Spectral flux (SF) measures the variability of the spectrum over time, i.e., SF is the average variation of the spectral magnitude between two adjacent signal segments. To compute the SF, the signal is divided to M segments each containing N samples. The SF between two adjacent signal segments X_m and X_{m-1} is[19],

$$SF(m) = \left[\sum_{k=1}^{N}\left(Y_m(k) - Y_{m-1}(k)\right)^P\right]^P,\ m = 1, 2, \ldots\ldots, M \tag{1}$$

In (1), $Y_m(k)$ is the absolute spectrum of the mth segment given by,

$$Y_m(k) = \left|\sum_{n=1}^{N} X_m(n) e^{\frac{-j2\pi kn}{N}}\right| \quad k = 1, 2 \ldots \ldots N \,\&\, m = 1, 2 \ldots \ldots M \tag{2}$$

The overall spectral flux of the signal is the average of SF between adjacent segments[20],

$$SF = \frac{1}{M-1}\sum_{m-1}^{M} SF(m) \tag{3}$$

In the current implementation of the spectral flux[20], the value of the exponent P in (1) is kept as 2. The signal segments of 1 second duration is preferable when the signal record is of sufficiently long duration.

We note that the current formulation of the spectral flux has limitations as it considers the disparity between the spectra of adjacent signal segments.

Spectral Spread

Spectral spread is the second central moment of the spectrum. The spectral spread is computed from the deviation of the spectrum from the spectral centroid[17] i.e., from the half-length magnitude spectrum of signals. The first half of the spectrum $Y(k),\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ of a signal comprising N samples corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the component frequencies, since the second half is just a replication of the first half. As described previously[17,21], the spectral spread is computed as,

$$Spectral\ Spread = \sqrt{\frac{\sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|\left(f(k) - f_{Centroid}\right)^2}{\sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|}} \tag{1}$$

Spectral Skewness

The spectral skewness measures the symmetry of the spectrum around the spectral centroid. Spectral skewness is the third central moment of the spectrum, and is computed from the half-length magnitude spectrum of signals. The first half of the spectrum $Y(k),\ k = 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ of a signal comprising N samples corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the

component frequencies, since the second half is just a replication of the first half. As described previously[21,22], the spectral skewness is computed as,

$$\frac{\sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|\left(f(k) - f_{Centroid}\right)^3}{(Spectral\ Spread)^3 \sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|} \quad (1)$$

Spectral Kurtosis

The spectral kurtosis measures the flatness or non-Gaussianity of the spectrum around its centroid. The spectral kurtosis is the fourth central moment of the spectrum, and is a parameter in frequency domain indicating how the impulsiveness of a signal varies with frequency. It is computed from the half-length magnitude spectrum of signals. The first half of the spectrum $Y(k),\ k\ =\ 1, 2, \ldots\ldots, \frac{[(N}{2)+1]}$ of a signal comprising N samples corresponding to the frequency bins from 0 to $\frac{f_s}{2}$ is enough to recognize the component frequencies, since the second half is just a replication of the first half. As described previously[21,22], spectral kurtosis is computed as,

$$\frac{\sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|\left(f(k) - f_{Centroid}\right)^4}{(Spectral\ Spread)^4 \sum_{k=1}^{\frac{[(N}{2)+1]}} |Y(k)|} \quad (1)$$

Complexity Measures

Lyapunov Exponent

The Lyapunov Exponent examines whether a stationary univariate time series is non-chaotic. It is often used to characterize the rate of separation of infinitesimally close trajectories in phase space to distinguish different attractors[23–26]. A negative Lyapunov Exponent indicates convergence, while positive Lyapunov Exponent demonstrates divergence and chaos. The magnitude of Lyapunov Exponent is an indicator of the rate of convergence or divergence of the infinitesimally close trajectories.

The Lyapunov Exponent function first generates a delayed reconstruction $Y_{1:N}$ with embedding dimension m, and lag τ. For each arbitrary point in the delayed reconstruction $Y_i,\ 1, 2, 3, \ldots\ldots, N$, define a nearest point Y_{i^*} based on two constraints[23]. Constraint 1 imposes that Euclidean distance

between the values Y_i and Y_{i^*} respectively at contextual point i and the neighbor point i^* is minimum, and the second constraint imposes that the neighbor point is outside the minimum separation limit D such that,

$$\|Y_i - Y_{i^*}\| \; and \; \left|i - i^*\right| > D \; given \; D = \frac{1}{f_m} \tag{1}$$

The minimum separation limit is considered as equal to the mean period which is the reciprocal of the mean frequency f_m. As described by Rosenstein et al[23]., The Lyapunov Exponent for a user-defined expansion range $\left[K_{min} K_{max}\right]$ is calculated as,

$$\lambda_i = \frac{1}{\left(K_{max} - K_{min} + 1\right)dt} \sum_{K=K_{min}}^{K_{max}} \frac{1}{K} \ln ln \left(\frac{\|Y_{i+k} - Y_{i^*+K}\|}{\|Y_i - Y_{i^*}\|}\right) \tag{2}$$

where dt is the sampling interval (reciprocal of sampling frequency). The term, $\ln ln \left(\frac{\|Y_{i+k} - Y_{i^*+K}\|}{\|Y_i - Y_{i^*}\|}\right)$ is known as logarithmic divergence. The variable K runs through finite number of steps with equal size.

The Lyapunov Exponent is computed from the linear regression fit among the step values in the $\left[K_{min} K_{max}\right]$ and the exponents λ_i, $1, 2, 3, \ldots\ldots, N$ obtained via a polynomial fit. In the practical implementation the constant term of the first-degree polynomial fit is taken as the Lyapunov Exponent.

The Phase-Space Reconstruction converts observed time series to state vectors. Given a uniformly sampled univariate time signal[27],

$$X_1 = [x_{11}, x_{12}, x_{13}, \ldots\ldots, x_{1N}]^T \tag{3}$$

where, N is the length of the time series. The subscript '1' indicate 'univariate'.

Phase-Space Reconstruction computes the delayed reconstruction as,

$$X_{1i}^r = \left(x_{1i}, x_{1(i+\tau_1)}, \ldots\ldots, x_{1\left(i+\left(m_1-1\right)\tau_1\right)}\right), \; i = 1, 2, \ldots\ldots N - \left(m_1 - 1\right)\tau_1 \tag{4}$$

where τ_1 is the lag, and m_1 is the embedding dimension for X_1.

Hurst Exponent

The Hurst Exponent is also referred to as the "index of dependence" or "index of long-range dependence"[28]. It quantifies the relative tendency of a time series either to regress strongly to the

mean or to cluster in a direction. Hurst Exponent in the range 0.5-1 indicates a time series with long-term positive autocorrelation, meaning that the decay in autocorrelation is slower than exponential, following a power law; for the series it means that a high value tends to be followed by another high value and that future excursions to more high values do occur. Hurst Exponent in the range 0 - 0.5 indicates a time series with long-term switching between high and low values in adjacent pairs, meaning that a single high value will probably be followed by a low value and that the value after that will tend to be high, with this tendency to switch between high and low values lasting a long time into the future, also following a power law. A value of Hurst Exponent equal to 0.5 indicates short memory, with (absolute) autocorrelations decaying exponentially quickly to zero. There are three different methods for estimating the Hurst Exponent; rescaled range (R/S) analysis, Detrended Fluctuation Analysis (DFA), and periodogram regression. R/S method of computing the Hurst Exponent is used here. The signal is divided into M segments with N samples in each segment such that $x_n^m,\ n = 1, 2, \ldots\ldots, N\ and\ m = 1, 2, \ldots\ldots, M$ represents an arbitrary sample in the mth segment X^m. Each signal segment X^m is normalized by subtracting the mean of the samples μ_m in it. An arbitrary sample in the mth normalized signal segment Y^m is[29],

$$y_n^m = x_n^m - \mu_m,\ y_n^m \in Y^m,\ n = 1, 2, \ldots\ldots, N\ and\ m = 1, 2, \ldots\ldots, M \tag{1}$$

From each normalized segment its cumulative sum is computed. The cumulative sum of the sample values in normalized segment Y_m is [29],

$$c_n^m = \sum_{i=1}^{n} y_n^m, given\ c_n^m \in C^m,\ n = 1, 2, \ldots\ldots, N\ and\ m = 1, 2, \ldots\ldots, M \tag{2}$$

In (2), C^m is the cumulative sum array corresponding to the normalized segment Y_m. From each cumulative sum array C^m, the range of values in the array R^m is computed as[2],

$$R^m = max\left\{c_1^m,\ c_2^m, \ldots\ldots, c_n^m, \ldots\ldots c_N^m\right\} - min\left\{c_1^m,\ c_2^m, \ldots\ldots, c_n^m, \ldots\ldots c_N^m\right\} \tag{3}$$

The range R^m of the m^{th}segment is divided with standard deviation of the sample values σ^m in the segment X^m to obtain the rescaled range $\frac{R^m}{\sigma^m}$. The mean of the rescaled range values from all segments are computed following this as [2],

$$\left[\frac{R}{\sigma}\right]_N = \frac{1}{M}\sum_{m=1}^{M}\frac{R^m}{\sigma^m} \tag{4}$$

In (4), the subscript $[.]_N$ denotes the dependency of the mean of the rescaled range values on the selection of the number of samples in the segment N. As described previously[29], the approximate relation [2],

$$\left[\frac{R}{\sigma}\right]_N \approx KN^H \Rightarrow \log log\left(\left[\frac{R}{\sigma}\right]_N\right) = log(K) + Hlog(N) \tag{5}$$

In (5), K is a constant. Simple way to estimate the Hurst Exponent H is to compute mean of the rescaled range values $\left[\frac{R}{\sigma}\right]_N$ for different selections of the length of the segment N and compute the slope via least square regression. If the signal is persistent then the slope is greater than 0.5. A persistent time series is one in which the value of the variable at any instant is closely related to the previous value. If the signal is anti-persistent, then the slope is less than 0.5. For white noise, ideally the slope is expected to be equal to 0.5. However, it should be noted that for small N there is a significant deviation from the 0.5 slope. To compensate this, theoretical $E\left[\frac{R}{\sigma}\right]_N$ for white noise is estimated and the values in the $\left[\frac{R}{\sigma}\right]_N$ is adjusted by subtracting $E\left[\frac{R}{\sigma}\right]_N$. The slope of first-degree polynomial fit between $\left[\frac{R}{\sigma}\right]_N - E\left[\frac{R}{\sigma}\right]_N$ plus 0.5 gives the compensated value of the Hurst Exponent. The generic formulation to compute the theoretical value of the rescaled range values $E\left[\frac{R}{\sigma}\right]_N$ for different lengths of the segments N of white noise is [2],

$$E\left[\frac{R}{\sigma}\right]_N = \begin{cases} \frac{\left[N-\left(\frac{1}{2}\right)\right]}{N} \frac{\Gamma\left[\frac{(N-1)}{2}\right]}{\sqrt{\pi}\Gamma\left[\frac{N}{2}\right]} \sum_{i=1}^{N-1} \sqrt{\frac{N-i}{i}} & N \leq 340 \\ \frac{\left[N-\left(\frac{1}{2}\right)\right]}{N} \frac{1}{\sqrt{\frac{N\pi}{2}}} \sum_{i=1}^{N-1} \sqrt{\frac{N-i}{i}} & Otherwise \end{cases} \tag{6}$$

In (6), $\Gamma(.)$ denotes the Euler gamma function.

Correlation Dimension

The Correlation Dimension (CD) measures dimensionality of the space occupied by a set of random points & is used to distinguish between deterministic chaos and random noise[30,31]. The CD value is estimated as the slope of the line fit between arbitrary number of discrete points within the range of radius of similarity and the corresponding correlation integral values. The value of CD is directly proportional to the level of chaos in the system, that is, a higher value of CD represents a high level of chaotic complexity in the system.

The first step of computing the CD is examining the range of radius of similarity $[R_{min}\ R_{max}]$. As described previously[30,31], often upper range of the radius of similarity R_{max} is computed from the covariance matrix C_Y of the reconstructed phase space Y as,

$$R_{max} = 0.2\sqrt{Tr(C_Y)} \tag{1}$$

In (1), the operator $Tr(.)$ symbolizes the trace operation that calculates the sum of the diagonal elements. In general,

$$Tr(A) = \sum_{i=1}^{M} a_{ii} = a_{11} + a_{22} + \ldots\ldots\ldots + a_{ii}\ldots\ldots + a_{MM} \quad given\ a_{ii} \in A \tag{2}$$

In (2), A is a dummy square metric with M rows and columns. The lower range of the radius of similarity is taken as $\frac{R_{max} = R_{max}}{1000}$. The covariance matrix C_Y has column variances of the reconstructed phase space along its diagonal. Following this, an array $R = \left\{r_1, r_2, \ldots\ldots, r_q, \ldots\ldots, r_Q\right\}$ that comprises exponents of finite number of linearly spaced discrete radii values between $[log(R_{min})\ log(R_{max})]$ is created. Q is the user-defined number of discrete radii values. We used the number of discrete radii values as 10. For each discrete radius value r_q, $q = 1, 2, \ldots\ldots, Q$, the total number of samples within that radius $\delta_i\left(r_q\right)$ at all points in phase space representation $\left\{Y = y_1, y_2, \ldots\ldots, y_i, \ldots\ldots y_N\right\}$ is computed as,

$$\delta_i\left(r_q\right) = \sum_{j=1}^{N} \psi\left(\|y_i - y_j\|\right) given\ j \neq i,\ i = 1, 2, \ldots\ldots, N \tag{3}$$

In (3), $\psi(.)$ is an indicator function given by [1-2],

$$\psi\left(\|y_i - y_j\|\right) = \{1\ if\ \|y_i - y_j\| > r_q\ 0\ Otherwise \tag{4}$$

In the next step, the correlation integral $Z\left(r_q\right)$, for each discrete radius value r_q, $q = 1, 2, \ldots\ldots, Q$, is computed as [1-2],

$$Z\left(r_q\right) = \frac{2}{N(N-1)} \sum_{i=1}^{N} \delta_i\left(r_q\right),\ \ q = 1, 2, \ldots\ldots, Q \tag{5}$$

The correlation integral estimates the mean probability that the states of a system are close at two different time intervals, which reflects self-similarity. The CD value is estimated as the slope of the line fit between $Z(r_q)$ and r_q.

Kolmogorov Complexity

Complexity of a binary sequence can be quantified by the way of the simplest computer program that can generate it[32,33] and expressed in terms of the Kolmogorov complexity[34]. The Kolmogorov complexity (Lempel-Ziv Complexity (LZC)) of a constantly sampled signal X is measured from a binary string B obtained via coarse granulation[35–37]. The binary string $B = b_1 b_2 b_3 \ldots\ldots b_i \ldots\ldots b_N$ will have number of bits that is equal to the number of samples N in the input signal X. The notion b_i, $i = 1, 2, \ldots\ldots N$ denotes an arbitrary bit in the binary string B. The Kolmogorov complexity is the total number of substrings present in the binary string[36,37] B. For computing the Kolmogorov complexity, its value is initialized as $C = 1$. For estimating the total number of substrings present, the binary string B is then split into two individual strings P and Q such that $P = b_1 b_2 b_3 \ldots\ldots b_i$ and $Q = b_{i+1}$. Following this, a vocabulary of substrings $V(R)$ present in a string R obtained by concatenating P and Q and excluding the last bit of the concatenated string PQ is formed. The vocabulary $V(R)$ is a set of all substrings present in the string R. In the next step, it is checked whether Q is a substring of $V(R)$. The condition $Q \in V(R)$ implies b_{i+1} is a substring of $b_1 b_2 b_3 \ldots\ldots b_i$. If the condition $Q \in V(R)$ is satisfied, the string P is kept intact and the string Q is updated by including the next bit such that $Q = b_{i+1} b_{i+2}$. The process of updating Q is recursed such that $Q = b_{i+1} b_{i+2} \ldots\ldots b_{i+j}$ until the condition $Q \notin V(R)$ that implies $b_{i+1} b_{i+2} \ldots\ldots b_{i+j}$ is not a substring of $b_1 b_2 \ldots\ldots b_i b_{i+1} b_{i+2} \ldots\ldots b_{i+j-1}$, is met. The complexity value C is incremented by one if the condition $Q \notin V(R)$ is met implying the availability of a new substring and the strings P and Q are updated respectively to $P = b_1 b_2 \ldots\ldots b_i b_{i+1} b_{i+2} \ldots\ldots b_{i+j}$ and $Q = b_{i+j+1}$. The above steps are repeated until $i + j < N$. The final value of the complexity is normalized by its possible upper limit to alleviate the bias caused by the number of samples in the signal such that[36–38],

$$c = \frac{C}{C_U} \; given \; C_U = \frac{N}{log_2(N)} \tag{1}$$

In (1), the variables c and C_U respectively denote the normalized Kolmogorov complexity estimate and its possible upper limit.

Katz's Fractal Dimension (KFD)

Greater KFDs indicate a higher complexity of signals. The KFD of a discrete signal X comprising N samples is computed from three terms[39–41] that are total length of the curve L, planar extent or diameter of the waveform d, and the number of steps in the waveform S as,

$$KFD(X) = \frac{log(S)}{\left[\log log\,(S) + log(\frac{d}{L})\right]} \tag{1}$$

Total length of the curve in the context of a signal is the sum of the distances between its successive samples. Total length of the curve L expressed as a sum of the Euclidean distances between successive samples is,

$$L = \sum_{n=1}^{N-1} \sqrt{[(n+1) - n]^2 + [x(n+1) - x(n)]^2} \tag{2}$$

In (2), the distances along horizontal and vertical axes of the two-dimensional plane are considered for computing the Euclidean distance between successive samples. The planar extent or diameter of the waveform d is the maximum of the distances of all samples in the signal from the very first sample $D(n),\ n = 2, 3, \ldots\ldots, N$as,

$$d = max\{D(n)\},\ n = 2, 3, \ldots\ldots, N\ Given\ D(n) = \sqrt{[n - 1]^2 + [x(n) - x(1)]^2} \tag{3}$$

The number of steps in the waveform S is the ratio of the total length of the curve L, and the average distance between its successive samples as,

$$S = \frac{\sum_{n=1}^{N-1} \sqrt{[(n+1)-n]^2 + [x(n+1)-x(n)]^2}}{\frac{1}{N-1} \sum_{n=1}^{N-1} \sqrt{[(n+1)-n]^2 + [x(n+1)-x(n)]^2}} = N - 1 \tag{4}$$

Hence,

$$KFD(X) = \frac{log(N-1)}{\left[\log log\,(N-1) + log(\frac{d}{L})\right]} \tag{5}$$

Higuchi's Fractal Dimension

The HFD (D) describes the degree of self-similarity along a 1D time series. It describes how fast (slope) the temporal derivatives of the signals constructed by extracting samples at regular intervals are changing with respect to the intervals. It is a bounded statistic with range[42,43] between 1 & 2. A higher D corresponds to higher complexity. The HFD value of a smooth curve (for example linear, a low frequency sine wave or less complex curve) is ≈1, while the HFD of random white noise or a more complex curve is ≈2.

For computing the Higuchi's Fractal Dimension (HFD)[42–44], we represent the discrete input signal sampled at a constant rate $X = \{x(1),\ x(2), \ldots\ldots, x(i) \ldots\ldots, x(N)\}$ as a set of reconstructed time series. The notion $x(i),\ i = 1, 2, \ldots\ldots, N$ indicate an arbitrary sample in the discrete signal comprising N samples. As described previously[42,43], the set of reconstructed time series is,

$$Y_k^m = x(m),\ x(m+k), x(m+2k), \ldots\ldots, \ldots\ldots, x\left(m + \lfloor \frac{N-m}{k} \rfloor k\right),\ m = 1, 2, \ldots\ldots, k\ \&\ k = 1, 2, \ldots\ldots K_{max} \tag{1}$$

In (1), k indicates the number of samples elapsed/interval between two consecutive samples in the reconstruction Y_k^m. The notion $\lfloor . \rfloor$ is the Gauss operator that extract the integer part of the argument (greatest integer that is less than or equal to the operator argument). In practical implementations of the HFD, K_{max} is often set as equal to $\frac{N}{2}$. For various values of k corresponding sets of reconstructions Y_k^m are computed and the curve lengths are computed[45]. The curve length L_k^m of an arbitrary reconstruction Y_k^m is,

$$L_k^m = \frac{\left[\left(\sum_{i=1}^{\lfloor \frac{N-m}{k} \rfloor} |x(m+ik) - x(m+(i-1)k)|\right)\left(\frac{(N-1)}{\lfloor \frac{N-m}{k} \rfloor k}\right)\right]}{k},\ m = 1, 2, \ldots\ldots, k\ \&\ k = 1, 2, \ldots\ldots K_{max} \tag{2}$$

The mean curve length L_k for a set of reconstructions $Y_k^m,\ m = 1, 2, \ldots\ldots, k$ is then computed from the individual curve lengths $L_k^m,\ m = 1, 2, \ldots\ldots, k$ as,

$$L_k = \frac{1}{k} \sum_{m=1}^{k} L_k^m,\ k = 1, 2, \ldots\ldots K_{max} \tag{3}$$

If $L_k \propto k^{-D}$, the curve is a fractal with dimension D. In practical implementations of HFD, a polynomial fit $log(L_k) = C_0 + C_1 log(\frac{1}{k})$ is obtained via least square regression. The slope of the line C_1 gives the value of the HFD. Alternatively[46], slope of the fit can be estimated as,

$$\frac{\left(p_k - \mu_P)\right)}{\left(\mu_k\right)^2)}, \; given\; \mu_k = \frac{1}{K_{max}} \sum_{k=1}^{K_{max}} k, \;\; \mu_P = \frac{1}{K_{max}} \sum_{k=1}^{K_{max}} p_k, \; and\; p_k = \log log\, \frac{1}{k}\,, k = 1, 2, ...K_{max} \tag{3}$$

Hjorth Parameters

Hjorth Activity, Hjorth Mobility, and Hjorth Complexity all together constitute the Hjorth parameters[47]. The Hjorth complexity reflects the degree of similarity of the test signal to a pure sine wave. It converges to 1 if the test signal is more like a pure sine wave.

$$Hjorth\; Activity = \frac{1}{N} \sum_{n=1}^{N} (x(n) - \bar{x})^2 = Var(x) \tag{1}$$

$$Hjorth\; Mobility = \sqrt{\frac{Var(\nabla_x)}{Var(x)}} \tag{2}$$

$$Hjorth\; Complexity = \frac{Mobility(\nabla_x)}{Mobility(x)} \tag{3}$$

Entropy

State Space Correlation Entropy (SSCE)

State Space Correlation Entropy (SSCE) is the Shannon entropy of values in the upper triangle of the covariance matrix of phase-space reconstruction[48,49]. If the signal has temporal self-similarity the values in the upper triangle of the covariance matrix of phase-space reconstruction will be high. If temporal self-similarity is very high in the signal, numbers of occurrences of values in the upper triangle of the covariance matrix tend to become uniform, its Shannon entropy and thereby the value of SSCE also tends to be high.

Approximate Entropy (ApEn)

Approximate Entropy (ApEn) measures the amount of irregularity and unpredictability of fluctuations in a signal. The value of ApEn is directly proportional to the irregularity, and thus, the ApEn of the

perfectly regular signal is expected to be significantly smaller than that of a random signal. In other words, a signal containing highly repetitive patterns will have a relatively small value of ApEn while a less predictable/deterministic signal will have a relatively larger value.

The first step of computing the ApEn is computing radius of similarity R [50]. It is computed from the covariance matrix C_Y of the reconstructed phase space Y of signal X as,

$$R = 0.2\sqrt{Tr(C_Y)} \tag{1}$$

In (1), the operator $Tr(.)$ symbolizes the trace operation that calculates the sum of the diagonal elements. In general,

$$Tr(A) = \sum_{i=1}^{M} a_{ii} = a_{11} + a_{22} + \ldots\ldots\ldots + a_{ii}\ldots\ldots + a_{MM} \;\; given\; a_{ii} \in A \tag{2}$$

In (2), A is a dummy square metric with M rows and columns. The covariance matrix C_Y has column variances of the reconstructed phase space along its diagonal. The total number of samples δ_i within radius of similarity R at all points $i = 1, 2, \ldots\ldots, N$ in phase space representation $\left\{Y = y_1, y_2, \ldots\ldots, y_i, \ldots\ldots y_N\right\}$ is computed as,

$$\delta_i = \sum_{j=1}^{N} \psi\left(\|y_i - y_j\|\right) given\; j \neq i,\; i = 1, 2, \ldots\ldots, N \tag{3}$$

In (3), N is the number of values in the phase space reconstruction and $\psi(.)$ is an indicator function given by,

$$\psi\left(\|y_i - y_j\|\right) = \{1\; if\; \|y_i - y_j\| > R\; 0\; Otherwise \tag{4}$$

Approximate Entropy (ApEn) of the signal X is,

$$\Phi(m) - \Phi(m+1)\; where\; \Phi(m) = \frac{1}{(N-m+1)} \sum_{i=1}^{N-m+1} \delta_i \tag{5}$$

In (5), m is the embedding dimension used during the phase space reconstruction. The idea is to compute average number of samples within radius of similarity $\Phi(m)$ for an embedding dimension m and $\Phi(m+1)$ for embedding dimension $m+1$ and find the difference $\Phi(m) - \Phi(m+1)$.

Permutation Entropy

Permutation Entropy measures the randomness of a signal based on the principles of symbol dynamics. It measures the uniformity of the histogram of symbols derived from small signal segments (snippets) of finite length (rank). In other words, Permutation Entropy quantifies the diversity of patterns that constitute a signal and is described previously[51,52]. Consider a discrete signal X with N number of samples in it.

$$X = \left\{x_1, x_2, x_3, \ldots\ldots, x_n, \ldots\ldots, x_N\right\} \tag{1}$$

In (1), x_n, $n = 1, 2, \ldots\ldots, N$ denotes an arbitrary sample. For a selection of order K, $N - (K - 1)$ sequences are generated from the signal X,

$$S = \left\{\left\{x_1, \ \ldots\ldots, x_{1+(K-1)}\right\}, \left\{x_2, \ \ldots\ldots, x_{2+(K-1)}\right\}, \ldots\ldots, \left\{x_n, \ \ldots\ldots, x_{n+(K-1)}\right\}, \ldots\ldots, \left\{x_{N-(K-1)}, \ \ldots\ldots, x_N\right\}\right\} \tag{2}$$

In (2), $S = \left\{s_1, s_2, \ldots\ldots, s_n, \ldots\ldots, s_{N-(K-1)}\right\}$ denotes the set of sequences and $s_n = \left\{x_n, \ \ldots\ldots, x_{n+(K-1)}\right\}$ represents an arbitrary sequence. Each sequence is encoded as a symbol of K decimal numbers within $[0 \ K - 1]$. The sequence s_n can also be written as,

$$s_n = \left\{x_{n+0}, \ \ldots\ldots, x_{n+k}, \ldots\ldots, x_{n+(K-1)}\right\} \tag{3}$$

In (3), x_{n+k}, $k = 0, 1, \ldots\ldots, K - 1$ denotes an arbitrary sample in the sequence s_n. For encoding a sequence as a symbol, the sequence is first sorted in ascending order and each element in the sorted sequence x_{n+k}, $k = 0, 1, \ldots\ldots, K - 1$ is replaced by k . For example, consider an arbitrary sequence $s_n = \{5, 4, 6\}$ with $K = 3$, $k = \{0, 1, 2\}$. In the sequence, $x_{n+0} = 5, x_{n+1} = 4, \ and \ x_{n+2} = 6$. After sorting the order of the elements in the sequence will be $\left\{x_{n+1}, x_{n+0}, x_{n+2}\right\}$ and the corresponding symbol will be $\{102\}$. Likewise, there could be $K!$ types of symbols for an order of K. Permutation entropy is computed from probabilities of occurrences of various symbol types in the whole encoded signal using the Shannon's formulation[51,52],

$$PE(X, K) = \ - \sum_{i=1}^{K!} P_i log_2(P_i) \tag{4}$$

In (4), P_i represents the relative frequencies of the possible patterns of symbol sequences, termed permutations. In order make the permutation entropy estimate unbiased against the selection of the order, there is an alternate formulation known as 'permutation entropy per symbol' given by,

$$PE(X,K) = \frac{-1}{K-1}\sum_{i=1}^{K!} P_i log_2(P_i) \quad (5)$$

Sample Entropy (SampEn)

Sample Entropy (SampEn) equals the negative average natural logarithm of the conditional probability that two signal segments with a finite number of samples that are similar will remain similar even if the segment length is increased by one more sample[53,54]. Higher the value of the SampEn higher will be complexity of the signal. To compute SampEn, a signal $X(n)$, $n = 1, 2, ..., N$ with N samples is converted into a two-dimensional representation by rearranging the samples in it. For an embedding dimension of m, the 2D representation will have $m + 1$ columns. An arbitrary value in the nth row of the ith column of the 2D representation is,

$$R_i(n) = X(i + n - 1), n = 1, 2, ..., (N + 1 - i), \ and\ i = 1, 2, ..., m + 1 \quad (1)$$

The 2D representation thus obtained will have N rows and $m + 1$ columns. During the practical realization of this logic, unequal number of values in each column make it difficult to stack the columns to form a 2D matrix. This issue is usually rectified by creating an initial matrix of non-integer values (NaN) and replacing the NaN values with samples from the signal at the desired coordinates. The first three rows of the 2D representation are,

$$R_1(1), R_2(1), ..., R_{m+1}(1) = X(1 + 1 - 1), X(2 + 1 - 1), ..., X(m + 1 + 1 - 1) = X(1), X(2), ..., X(m +$$

$$(m + 1 + 2 - 1) = X(2), X(3), ..., X(m + 2) \quad (3)$$

$$(m + 1 + 3 - 1) = X(3), X(4), ..., X(m + 3) \quad (4)$$

Likewise, an arbitrary row in the 2D representation is,

$$(m + 1 + n - 1) = X(n), X(n + 1), ..., X(m + n) \quad (5)$$

Thus, three arbitrary consecutive rows in the 2D representation can be written as,

$$- 1)\, X(n) \,.....\, X(m + n - 1)\, X(n)\, X(n + 1) \,.....\, X(m + n)\, X(n + 1)\, X(n + 2) \,.....\, X(m + n + 1) \quad (6)$$

In the next step of a one-dimensional distance array that comprises the distance measures between each possible combinations of rows (observations) of the 2D representation is computed. As the 2D representation matrix has N rows, $\frac{N(N-1)}{2}$ different combinations are possible. Consequently, the one-dimensional array of distance measures will have $\frac{N(N-1)}{2}$ elements in it. Previously, Chebychev distance is used[54–57]. The Chebychev distance $d(R(n), R(n - 1))$ between two rows $R(n) = R_1(n), R_2(n), ..., R_{m+1}(n)$ and $R(n - 1) = R_1(n - 1), R_2(n - 1), ..., R_{m+1}(n - 1)$ is,

$$d(R(n), R(n-1)) = \left\{\left|R_j(n) - R_j(n-1)\right|\right\}, \quad j = 1, 2, \ldots\ldots, m+1 \tag{7}$$

Theoretically, the Chebychev distance between two rows is the highest of the absolute differences between coinciding sets of elements. In the next step, another distance array is computed from the 2D representation of the signal with an embedding dimension of $m - 1$. For an embedding dimension of $m - 1$, the 2D representation of the signal will have m columns. Instead of computing the 2D representation again, the first columns of the previous 2D representation can be extracted. Following this, the total numbers of values less than the threshold is computed from both distance arrays. The total number of values C in a distance array d that are less than the threshold T is,

$$C = \sum_{k=1}^{\frac{N(N-1)}{2}} \Psi(T - d(k)) \; where \; \Psi(T - d(k)) = \{1 \; if \; T - d(k) > 0 \;\; 0 \; Otherwise \; and \; T = t\sigma_X \tag{8}$$

In (8), $\Psi(.)$ is a Heaviside function. The threshold T is the product of the user-defined tolerance t and standard deviation of the signal σ_X. The sample entropy of an input signal X is,

$$SampEn(X) = -\ln\left(\frac{C_{m+1}}{C_m + \varepsilon}\right)\left(\frac{N-m+1}{N-m-1}\right) \tag{9}$$

In (9), C_{m+1} and C_m are total numbers of values less than the threshold computed respectively from distance arrays corresponding to embedding dimensions equal to $m + 1$ and m. The variable ε is a negligibly small number less than 2.2204×10^{-16} that is used to avoid division by zero and further indeterminacy. The first term of SampEn in (9) reflects, how the total numbers of temporally similar signal segments vary with respect to the segment length. If a small increase in the segment length (one sample) do not vary the number of temporally similar segments, the term will be equal to 1 and the natural logarithm will produce 0 as demonstrated in (10).

$$\left.\right) < 0 \; and \; C_{m+1} < C_m \Rightarrow -\ln\left(\frac{C_{m+1}}{C_m + \varepsilon}\right) > 0 \tag{10}$$

In a nonstationary signal, increasing the number of samples in the signal segments will reduce the total number of similar segments in the signal. Thus, in non-stationary signals $C_{m+1} < C_m$ and the value of SampEn will be high. In the extreme case, especially when the embedding dimension is high and signal is quite random, the total number of segments with mutual distance within the tolerance may tend to be zero as,

$$C_{m+1} = 0 \Rightarrow -\ln\left(\frac{C_{m+1}}{C_m + \varepsilon}\right) = -\ln(0) = \infty \tag{11}$$

Power Ratios

Four well-established power ratios are used here. They are Theta-Beta (TB) Ratio[58–60], Theta-Alpha (TA) Ratio[60–62], Gamma-Alpha (GA) Ratio and Alpha-Beta to Delta-Theta (ABDT) Ratio[63].

$$TB\ Ratio = \frac{P_\theta}{P_\beta},$$

$$TA\ Ratio = \frac{P_\theta}{P_\alpha}, GA\ Ratio = \frac{P_\gamma}{P_\alpha},$$

$$and\ ABDT\ Ratio = \frac{P_\alpha + P_\beta}{P_\delta + P_\theta} \qquad (1)$$

In (1), P_δ,P_θ, P_α , and $P_\theta$are total powers of the respective bands. The bandwidth considered are; $\delta: 1 - 4\ Hz$, $\theta: 5 - 8\ Hz$, $\alpha: 9 - 13\ Hz$, $\beta: 14: 30\ Hz$ and γ: 31:48 Hz.

Wavelet features

A total of six features such as approximation energy, detail energy, ratio of the approximation and detail energies termed relative wavelet energy, computed from the sub-bands obtained from single-level 1-D discrete wavelet transform decomposition and 1-D wavelet packet decomposition fall under the category of Wavelet features. From the 1-D wavelet decomposition, the percentage of energy corresponding to the approximation, and the percentages of energies corresponding to the details are computed. Similarly, percentages of approximation and details energies corresponding to the terminal nodes of the wavelet packet tree obtained via the wavelet packet decomposition are computed. Daubechies wavelet[64,65] of order 1 is used as the basis function for decomposition.

Pairwise Phase Connectivity

Pairwise Phase Consistency (PPC), introduced by Vinck et al. (2010)[66], represents an advancement in functional connectivity (FC) assessment and focuses on quantifying phase synchronization through coherency. Unlike traditional methods, PPC given in equation (1) remains unbiased by variable sample sizes by employing vector dot product computation rather than vector addition for oscillatory phase pairs. This measure is computed across frequencies ranging from 1 to

48 Hz with a granularity of 1 Hz. The results emphasize only statistically significant electrode pairs after FDR correction[67], with non-significant pairs assigned zero values.

$$PPC(f) = \frac{2}{N(N-1)} \sum_{j=1}^{N-1} \sum_{k=j+1}^{N} cos(\theta_j(f) - \emptyset_k(f)) \qquad (1)$$

Where N is the number of trials, θjf and $\emptyset kf$ are the relative phases of the two signals, in our case, different electrodes and f denote the frequency.

Functional connectivity (FC) analysis is done using custom MATLAB (Mathworks. Inc) codes and using the Fieldtrip Toolboox[68] with trial length of 1-second.

Neural Network & SHAP analysis

Power Spectral Density (PSD)

The PSD data from preprocessed EEG data via artifact subspace reconstruction (ASR) pipeline (Eyes Closed case, 30 channels from each subject) was used to train a Neural Network to classify between Alzheimer's Disease and Healthy Elderly categories. PSD was extracted using multi-taper method with single slepian taper using 'mtspectrum' function of Chronux toolbox[69]. 60 seconds of data is considered with 1 second epoch for PSD extraction with a frequency resolution of 1 Hz. 1-48 Hz frequency bins resulting in a feature vector length of 48 are considered for the classification using neural network. The time series data was then transformed into PSD data, followed by averaging all PSDs along the time axis. The dataset from Cohort1 and age-matched one from Cohort2 was used to train the model which does binary classification task of corresponding PSDs. Cohort1 comprises of 46 subjects in the Healthy Elderly (HE) category, and 46 subjects in the Alzheimer's Disease (AD) category. Each subject will contribute to 30 channels of data which corresponds to 30 PSDs. For training and testing the neural network, 1500 samples from each category were utilized. Rest of the data (300 samples from HE and 270 samples from AD) was kept as an unseen set to verify the performance of neural network. The age-matched PSD samples from dataset Cohort3 comprises of 30 subjects respectively in Healthy (HC) and Mild cognitive impairment (MCI) cases.

Neural Network Architecture (NN1) for PSD classification

The NN architecture is a sequential model with two hidden layers:

1. The initial layer is a 48 node layer as it represents all the binned frequencies (1 to 48 Hz) vector.

2. Dense Layer (64 neurons): This layer serves as the input layer with 3136 parameters (from respective weights and biases) and utilizes rectified linear unit (ReLU) activation.
3. Dropout Layer (Dropout rate of 0.5): This layer helps prevent overfitting by randomly dropping out 50% of the neurons during training.
4. Dense Layer (32 neurons): This layer, with 2080 parameters, also employs ReLU activation.
5. Dropout Layer: Another dropout layer with a dropout rate of 0.5.
6. Final Dense Layer (2 neurons): The final layer, with 66 parameters, produces the output with two neurons representing the classification categories. It employs a softmax activation function for the classification task.

   See Figure 1 for the illustration of the proposed ANN architecture.

Model Summary of NN1

The total number of parameters in the model is 5,282, all of which are trainable. The model was trained on the PSDs from preprocessed EEG data from both HE and AD subjects using the specified architecture.

The neural network model was implemented using the TensorFlow Python package[70] , leveraging its capabilities for efficient deep learning model development and training. The neural network model made use of Adam optimizer (with a learning rate 0.001) and binary cross-entropy loss function. The model was then trained for 1000 epochs with a batch size of 16. It achieved a validation accuracy of 97.73% in classifying AD versus HE classes. For the age-matched Cohort2 dataset, the validation accuracy was 78.44% (HC versus MCI). This methodology allowed for the training of a neural network for classifying EEG based PSD data from individuals with and without Alzheimer's disease, providing a foundation for further analysis and interpretation of the results.

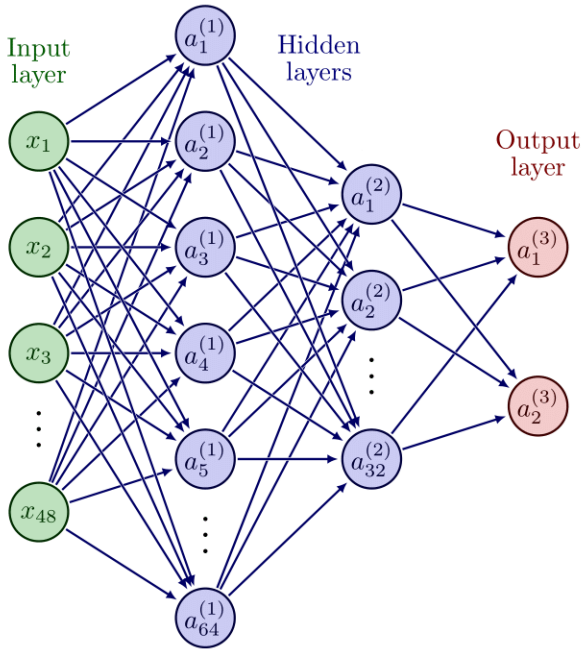

Figure 1: An illustration of the Neural Network architecture as described in Methods section.

SHapley Additive exPlanations (SHAP)

Feature importance (separability metrics) measures the extent to which a feature or variable influences the prediction results of a given model. A predictive model often uses multiple features, each contributing uniquely to the prediction. A model can be better understood or explained/interpreted when one knows the influence of each feature on the model's prediction accuracy. SHapley Additive exPlanations (SHAP) and permutation feature importance measures are two methods to quantify individual features' influence on a given model's prediction accuracy. Permutation importance is defined as the mean decrease in accuracy of the trained model when each feature is permuted. Saarela & Jauhiainen (2021)[71] have shown that there is no correspondence between the determinant features identified via linear (logistic regression with L1 penalization) and non-linear (Permutation) methods. The feature importance measured with the help of permutation method and SHAP are model-specific (depends on the classifier used). The 'feature importance' does not necessarily reflect the separability offered by the individual features in a binary classification paradigm. Instead, it reflects the fractional contribution of a feature towards accurate decision-making when multiple features drive the classifier.

Features with p-values much lower than the level of significance in the Wilcoxon rank sum test may have higher importance than the significant features in SHAP and Permutation method[71] (For proof, please refer to Table 2, Sean et al., 2022[72]

SHAP Analysis of NN1

SHAP (SHapley Additive exPlanations) is a game-theory based method employed to explain the predictions generated by any machine learning model. SHAP[73] values are model agnostic, which implies it can be used to interpret any ML model. Here we used SHAP for explaining our ANN model. We want to know which features (PSD frequency bins from 1 to 48 Hz) contribute most to the accuracy of the ANN. This analysis was done in Python package called SHAP.

Features from Cohort1 and Cohort3

The input data here comprises of 41 features extracted from datasets Cohort1 and Cohort3. We use a subset of samples for Cohort1 (46) and Cohort3 (30) after the process of age-matching. Each sample corresponds to 30 electrode data points. The 41 features are Variance, Skewness, Kurtosis, Shape Factor, Peak Amplitude, Impulse factor, Crest factor, clearance factor, Willison Amp, zero crossing rate, Sample Entropy, Approximate Entropy, permutation entropy, state space correlation entropy, correlation dimension, Higuchi Fractal dimension, Katz Fractal dim, Lyapunov Exponent, Hurst Exponent, Lempel-Ziv Complexity, Hjorth Activity, Hjorth Mobility, Hjorth Complexity, alpha dominant frequency, gamma dominant frequency, Spectral Roll Off, Spectral Centroid, Spectral Spread, Spectral Flux, Spectral Skewness, Spectral Kurtosis, Theta Beta Ratio, Theta Alpha Ratio, Gamma alpha ratio, Spectral Power Ratio, Wavelet Energy Approximate, Wavelet Energy Detail, Relative Wavelet Energy, Wavelet Packet Energy Approximate, Wavelet Packet Energy Detail, and Relative Wavelet Packet Energy.

Neural Network Architecture (NN2) for features

The NN architecture is a sequential model with two hidden layers:

1. The initial layer is a 41 node layer as it represents all the 41 features vector.
2. Dense Layer (36 neurons): This layer serves as the input layer with 1512 parameters (from respective weights and biases) and utilizes rectified linear unit (ReLU) activation.
3. Dense Layer (18 neurons): This layer, with 666 parameters, also employs ReLU activation.
4. Final Dense Layer (2 neurons): The final layer, with 38 parameters, produces the output with two neurons representing the classification categories. It employs a softmax activation function for the classification task.

Model Summary of NN2

The total number of parameters in the model is 2216, all of which are trainable. We ensembled 20 such individual models to get a final robust model by averaging weights of the individual models. Other than architecture, the parameters like learning rate, batch size, and number of epochs etc were kept same as in the case of first neural network NN1. Additionally, we made use of early stopping technique to avoid over-fitting.

SHAP Analysis of NN2

SHAP analysis was done on this ensemble neural network model too. Our aim was to see the most significantly contributing features in our classifier model.

Spearman's rank correlation

Spearman rank correlation[74][74] was used to test the degree of correlation between 41 different EEG features and MMSE scores across all electrodes and for all cohorts. The feature values for a specific feature corresponding to one of the electrodes were taken and the correlation was computed between these values and the corresponding MMSE scores of each AD or prodromal AD cohort. This process was repeated for all possible combinations of electrodes and features. The resulting correlation coefficient values were organised electrode-wise and were represented using a heat plot. This analysis provided the relationship between the EEG features and MMSE scores. The significance level was set at $p < 0.05$ to indicate statistical significance.

Wilcoxon rank sum test and false discovery rate (FDR) correction

The difference between the EEG features, PSD and functional connectivity across different groups (AD vs HC or prodromal AD vs HC) were tested using the non-parametric based Wilcoxon rank sum test[75,76][75,76]. This test was adopted in the study as it does not assume known distributions and hence does not deal with parameters such as variance and mean and is based on the median difference. For PSD analysis the test was carried out for each of the frequency bins (i.e., every 1 Hz resolution) and functional connectivity results were tested for each of the connections.

The multiple comparison was done to control the false discovery rate (FDR) using the procedure proposed by Benjamin and Hochberg[67][76]. The FDR correction was carried out across electrodes for EEG features and PSD analysis. But for functional connectivity analysis, FDR correction was carried out across frequency-wise connectivity matrices i.e., 48 points as 1-48Hz was considered for the analysis.

Normalized Features

The normalized EEG features shown in Figures 2C and 4C, were calculated first by averaging the features' values in the control group and then comparing each subject's value (subject 1 (n)/control group mean) to the control group mean.

GUI - Feature Analyzer 1.0

A standalone software application with a graphical user interface (GUI) was developed using MATLAB App Designer. This GUI-based software, termed 'Feature Analyzer 1.0', is designed to operate independently without needing any third-party software, including MATLAB, for feature extraction functions. Therefore, installation of MATLAB is not required to utilize this software. The Feature Analyzer 1.0 installation requires ~6.8 GB of space on the system at the time of installation and an internet. In addition, optimal performance is achieved on a PC or workstation with a minimum of 16 GB RAM, and the Feature Analyzer 1.0 is compatible with Windows operating systems.

The software features three main tabs: Feature Extraction, Result, and Developer Info. Users can manually or automatically extract features from input signals in the Feature Extraction tab, with corresponding results displayed in the Result tab. This tab includes a command window, output window, and a provision for visualizing imported data. Notably, the Feature Analyzer 1.0 directly extracts features from input signals without pre-processing, ensuring consistency and homogeneity in analysis protocols across users. Users are required to pre-process signals (e.g., denoising, artifact removal) before feature extraction using the software.

The Feature Analyzer 1.0 offers options to extract individual features from a list of 41 predefined features or execute all features sequentially using the "Execute All" button. Results are displayed in a new tab named Results and can be copied to the clipboard or automatically saved as a .csv file. The software includes a 'Help' button, which serves as a valuable resource for users by providing a comprehensive user manual. This manual is designed to facilitate smooth navigation and operation of the software, offering detailed instructions, guidelines, and tips for optimal utilization. By accessing the user manual through the 'Help' button, users can effectively troubleshoot issues, understand functionalities, and maximize the utility of the software.

To extract feature values for a signal, users must follow these steps:

1. Import the prior preprocessed signal. The GUI currently supports .csv, .txt, .edf, and .mat formats for the importing data.
2. Input the sampling frequency of the signal in Hz. Please note a minimum sampling frequency of 125 Hz is recommended.
3. Choose the feature from table of 41 – either automate or one at a time.
4. After the feature value is calculated, users can copy the feature values or save the results as a .csv file.

We note that a few features, such as HFD (Higuchi's Fractal Dimension) and SSCE (State Space Correlation Entropy), require a higher memory for processing longer signal lengths and larger data. In some cases, such features may not be extracted due to the system's limitations, not the Feature Analyzer 1.0's.

Supplementary References

[1] Güntekin, B. *et al.* (2023). Alterations of resting-state Gamma frequency characteristics in aging and Alzheimer's disease. *Cognitive Neurodynamics*. https://doi.org/10.1007/s11571-022-09873-4.

[2] Şeker, M. *et al.* (2021). Complexity of EEG Dynamics for Early Diagnosis of Alzheimer's Disease Using Permutation Entropy Neuromarker. *Computer Methods and Programs in Biomedicine*. https://doi.org/10.1016/j.cmpb.2021.106116.

[3] McKhann, G.M. *et al.* (2011). The diagnosis of dementia due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's and Dementia*. https://doi.org/10.1016/j.jalz.2011.03.005.

[4] Segal, D.L. (2010). Diagnostic and Statistical Manual of Mental Disorders, in *The Corsini Encyclopedia of Psychology*, Wiley, pp. 1–3.

[5] Tzimourta, K.D. *et al.* (2019). Analysis of electroencephalographic signals complexity regarding Alzheimer's Disease. *Computers & Electrical Engineering*. https://doi.org/10.1016/j.compeleceng.2019.03.018.

[6] Şeker, M. *et al.* (2021). Complexity of EEG Dynamics for Early Diagnosis of Alzheimer's Disease Using Permutation Entropy Neuromarker. *Computer Methods and Programs in Biomedicine*. https://doi.org/10.1016/j.cmpb.2021.106116.

[7] Albert, M.S. *et al.* (2011). The diagnosis of mild cognitive impairment due to Alzheimer's disease: Recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease. *Alzheimer's & Dementia*. https://doi.org/10.1016/j.jalz.2011.03.008.

[8] Bokil, H. *et al.* (2010). Chronux: A platform for analyzing neural signals. *Journal of Neuroscience Methods*. https://doi.org/10.1016/j.jneumeth.2010.06.020.

[9] Liu, H. *et al.* (2004). On-line outlier detection and data cleaning. *Computers & Chemical Engineering*. https://doi.org/10.1016/j.compchemeng.2004.01.009.

[10] Dagar, M. *et al.* (2018). Performance Comparison of Hampel and Median Filters in Removing Deep Brain Stimulation Artifact, pp. 17–28.

[11] Allen, D.P. *et al.* (2010). Suppression of deep brain stimulation artifacts from the electroencephalogram by frequency-domain Hampel filtering. *Clinical Neurophysiology*. https://doi.org/10.1016/j.clinph.2010.02.156.

[12] Kothe (2015). Artifact removal technique with signal reconstruction. issued 2015.

[13] Tăuţan, A.-M. *et al.* (2023). TMS-EEG perturbation biomarkers for Alzheimer's disease patients classification. *Scientific Reports*. https://doi.org/10.1038/s41598-022-22978-4.

[14] Melek, M. *et al.* (2021). An automatic EEG-based sleep staging system with introducing NAoSP and NAoGP as new metrics for sleep staging systems. *Cognitive Neurodynamics*. https://doi.org/10.1007/s11571-020-09641-2.

[15] Willison (1963). *J.Physiology*.

[16] Staudinger, T. and Polikar, R. (2011). Analysis of complexity based EEG features for the diagnosis of Alzheimer's disease. *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. https://doi.org/10.1109/IEMBS.2011.6090374.

[17] Giannakopoulos, T. and Pikrakis, A. (2014). Audio Features, in *Introduction to Audio Analysis*, Elsevier, pp. 59–103.

[18] Choi, J. *et al.* (2019). Resting-state prefrontal EEG biomarkers in correlation with MMSE scores in elderly individuals. *Scientific Reports*. https://doi.org/10.1038/s41598-019-46789-2.

[19] Scheirer, E. and Slaney, M.CONSTRUCTION AND EVALUATION OF A ROBUST MULTIFEATURE SPEECH/MUSIC DISCRIMINATOR.

[20] Scheirer, E.D. and M.S. "Construction and evaluation of a robust multifeature speech/music discriminator." 1997 IEEE International Conference on Acoustics, Speech, and Signal Processing 2 (1997): 1331-1334 vol.2.

[21] Geoffroy Peeters (2004). A large set of audio features for sound description (similarity and classification) in the CUIDADO project. *CUIDADO 1st Project Report*.

[22] Sutawanir (2017). Estimation of spectral kurtosis. *AIP Conference Proceedings*. https://doi.org/10.1063/1.4979440.

[23] Rosenstein, M.T. *et al.* (1993). A practical method for calculating largest Lyapunov exponents from small data sets. *Physica D: Nonlinear Phenomena*. https://doi.org/10.1016/0167-2789(93)90009-P.

[24] Jeong, J. *et al.* (1998). Non-linear dynamical analysis of the EEG in Alzheimer's disease with optimal embedding dimension. *Electroencephalography and Clinical Neurophysiology*. https://doi.org/10.1016/S0013-4694(97)00079-5.

[25] Eric W. Weisstein (2021). .

[26] Ma, M.K.-H. *et al.* (2021). Regularity and randomness in ageing: Differences in resting-state EEG complexity measured by largest Lyapunov exponent. *Neuroimage: Reports*. https://doi.org/10.1016/j.ynirp.2021.100054.

[27] B. Kliková and A. Raidl (2011). Reconstruction of Phase Space of Dynamical Systems Using Method of Time Delay. *Proceedings of the 20th Annual Conference of Doctoral Students*.

[28] Hurst, H.E. (1956). THE PROBLEM OF LONG-TERM STORAGE IN RESERVOIRS. *International Association of Scientific Hydrology. Bulletin*. https://doi.org/10.1080/02626665609493644.

[29] Weron, R. (2002). Estimating long-range dependence: finite sample properties and confidence intervals. *Physica A: Statistical Mechanics and its Applications*. https://doi.org/10.1016/S0378-4371(02)00961-5.

[30] Caesarendra, W. *et al.* (2013). An application of nonlinear feature extraction - A case study for low speed slewing bearing condition monitoring and prognosis. *2013 IEEE/ASME International Conference on Advanced Intelligent Mechatronics*. https://doi.org/10.1109/AIM.2013.6584344.

[31] Theiler, J. (1987). Efficient algorithm for estimating the correlation dimension from a set of discrete points. *Physical Review A*. https://doi.org/10.1103/PhysRevA.36.4456.

[32] Lempel, A. and Ziv, J. (1976). On the Complexity of Finite Sequences. *IEEE Transactions on Information Theory*. https://doi.org/10.1109/TIT.1976.1055501.

[33] Kaspar, F. and Schuster, H.G. (1987). Easily calculable measure for the complexity of spatiotemporal patterns. *Physical Review A*. https://doi.org/10.1103/PhysRevA.36.842.

[34] Kolmogorov, A.N. (1968). Three approaches to the quantitative definition of information. *International Journal of Computer Mathematics*. https://doi.org/10.1080/00207166808803030.

[35] Aboy, M. *et al.* (2006). Interpretation of the Lempel-Ziv Complexity Measure in the Context of Biomedical Signal Analysis. *IEEE Transactions on Biomedical Engineering*. https://doi.org/10.1109/TBME.2006.883696.

[36] Xu-Sheng Zhang and Roy, R.J. (2001). Derived fuzzy knowledge model for estimating the depth of anesthesia. *IEEE Transactions on Biomedical Engineering*. https://doi.org/10.1109/10.914794.

[37] Xu-Sheng Zhang *et al.* (1999). Detecting ventricular tachycardia and fibrillation by complexity measure. *IEEE Transactions on Biomedical Engineering*. https://doi.org/10.1109/10.759055.

[38] Jing Hu *et al.* (2006). Analysis of Biomedical Signals by the Lempel-Ziv Complexity: the Effect of Finite Data Size. *IEEE Transactions on Biomedical Engineering*. https://doi.org/10.1109/TBME.2006.883825.

[39] Katz, M.J. (1988). Fractals and the analysis of waveforms. *Computers in Biology and Medicine*. https://doi.org/10.1016/0010-4825(88)90041-8.

[40] Jesús Monge-Álvarez (2023). .

[41] Mohd Radzi, S.S. *et al.* (2019). Fractal Dimension and Power Spectrum of Electroencephalography Signals of Sleep Inertia State. *IEEE Access*. https://doi.org/10.1109/ACCESS.2019.2960852.

[42] Mehrabbeik, M. *et al.* (2022). Distinctive nonlinear dimensionality of neural spiking activity in extrastriate cortex during spatial working memory; a Higuchi fractal analysis. *Chaos, Solitons & Fractals*. https://doi.org/10.1016/j.chaos.2022.112051.

[43] Higuchi, T. (1988). Approach to an irregular time series on the basis of the fractal theory. *Physica D: Nonlinear Phenomena*. https://doi.org/10.1016/0167-2789(88)90081-4.

[44] Kesić, S. and Spasić, S.Z. (2016). Application of Higuchi's fractal dimension from basic to clinical neurophysiology: A review. *Computer Methods and Programs in Biomedicine*. https://doi.org/10.1016/j.cmpb.2016.05.014.

[45] Yahyaei, R. and Esat Özkurt, T. (2022). Mean curve length: An efficient feature for brainwave biometrics. *Biomedical Signal Processing and Control*. https://doi.org/10.1016/j.bspc.2022.103664.

[46] Liehr, L. and Massopust, P. (2020). On the mathematical validity of the Higuchi method. *Physica D: Nonlinear Phenomena*. https://doi.org/10.1016/j.physd.2019.132265.

[47] Hjorth, B. (1970). EEG analysis based on time domain properties. *Electroencephalography and Clinical Neurophysiology*. https://doi.org/10.1016/0013-4694(70)90143-4.

[48] Tripathy, R.K. *et al.* (2017). Analysis of physiological signals using state space correlation entropy. *Healthcare Technology Letters*. https://doi.org/10.1049/htl.2016.0065.

[49] Guo, X. *et al.* (2022). Emotional Activity Is Negatively Associated With Cognitive Load in Multimedia Learning: A Case Study With EEG Signals. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2022.889427.

[50] Pincus, S.M. (1991). Approximate entropy as a measure of system complexity. *Proceedings of the National Academy of Sciences*. https://doi.org/10.1073/pnas.88.6.2297.

[51] Bandt, C. and Pompe, B. (2002). Permutation Entropy: A Natural Complexity Measure for Time Series. *Physical Review Letters*. https://doi.org/10.1103/PhysRevLett.88.174102.

[52] Riedl, M. *et al.* (2013). Practical considerations of permutation entropy. *The European Physical Journal Special Topics*. https://doi.org/10.1140/epjst/e2013-01862-7.

[53] Richman, J.S. and Moorman, J.R. (2000). Physiological time-series analysis using approximate entropy and sample entropy. *American Journal of Physiology-Heart and Circulatory Physiology*. https://doi.org/10.1152/ajpheart.2000.278.6.H2039.

[54] Lord, B. and Allen, J.J.B. (2023). Evaluating EEG complexity metrics as biomarkers for depression. *Psychophysiology*. https://doi.org/10.1111/psyp.14274.

[55] Abásolo, D. *et al.* (2006). Entropy analysis of the EEG background activity in Alzheimer's disease patients. *Physiological Measurement*. https://doi.org/10.1088/0967-3334/27/3/003.

[56] Fang, J. *et al.* (2023). Quantification of transient evoked otoacoustic emissions by sample entropy to assess cochlear symptoms (Hearing and Tinnitus) in Meniere's disease. *Applied Acoustics*. https://doi.org/10.1016/j.apacoust.2023.109587.

[57] Al-Angari, H.M. and Sahakian, A. V. (2007). Use of Sample Entropy Approach to Study Heart Rate Variability in Obstructive Sleep Apnea Syndrome. *IEEE Transactions on Biomedical Engineering*. https://doi.org/10.1109/TBME.2006.889772.

[58] Azami, H. *et al.* (2023). Beta to theta power ratio in EEG periodic components as a potential biomarker in mild cognitive impairment and Alzheimer's dementia. *Alzheimer's Research & Therapy*. https://doi.org/10.1186/s13195-023-01280-z.

[59] Miao, Y. *et al.* (2021). Dynamic theta/beta ratio of clinical EEG in Alzheimer's disease. *Journal of Neuroscience Methods*. https://doi.org/10.1016/j.jneumeth.2021.109219.

[60] Baik, K. *et al.* (2022). Implication of EEG theta/alpha and theta/beta ratio in Alzheimer's and Lewy body disease. *Scientific Reports*. https://doi.org/10.1038/s41598-022-21951-5.

[61] Schmidt, M.T. *et al.* (2013). Index of Alpha/Theta Ratio of the Electroencephalogram: A New Marker for Alzheimer's Disease. *Frontiers in Aging Neuroscience*. https://doi.org/10.3389/fnagi.2013.00060.

[62] Özbek, Y. *et al.* (2021). Resting-state EEG alpha/theta power ratio discriminates early-onset Alzheimer's disease from healthy controls. *Clinical Neurophysiology*. https://doi.org/10.1016/j.clinph.2021.05.012.

[63] Flores-Sandoval, A.A. *et al.* (2023). Spectral power ratio as a measure of EEG changes in mild cognitive impairment due to Alzheimer's disease: a case-control study. *Neurobiology of Aging*. https://doi.org/10.1016/j.neurobiolaging.2023.05.010.

[64] AlSharabi, K. *et al.* (2022). EEG Signal Processing for Alzheimer's Disorders Using Discrete Wavelet Transform and Machine Learning Approaches. *IEEE Access*. https://doi.org/10.1109/ACCESS.2022.3198988.

[65] Vicchietti, M.L. *et al.* (2023). Computational methods of EEG signals analysis for Alzheimer's disease classification. *Scientific Reports*. https://doi.org/10.1038/s41598-023-32664-8.

[66] Vinck, M. *et al.* (2010). The pairwise phase consistency: A bias-free measure of rhythmic neuronal synchronization. *NeuroImage*. https://doi.org/10.1016/j.neuroimage.2010.01.073.

[67] Benjamini, Y. and Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B (Methodological)*. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.

[68] Oostenveld, R. *et al.* (2011). FieldTrip: Open Source Software for Advanced Analysis of MEG, EEG, and Invasive Electrophysiological Data. *Computational Intelligence and Neuroscience*. https://doi.org/10.1155/2011/156869.

[69] Bokil, H. *et al.* (2010). Chronux: A platform for analyzing neural signals. *Journal of Neuroscience Methods*. https://doi.org/10.1016/j.jneumeth.2010.06.020.

[70] TensorFlow Developers. (2023). (2023). TensorFlow (v2.15.0). Zenodo. .

[71] Saarela, M. and Jauhiainen, S. (2021). Comparison of feature importance measures as explanations for classification models. *SN Applied Sciences*. https://doi.org/10.1007/s42452-021-04148-9.

[72] Yu, S.C. *et al.* (2022). Sepsis Prediction for the General Ward Setting. *Frontiers in Digital Health*. https://doi.org/10.3389/fdgth.2022.848599.

[73] Lundberg, S.M., & L.S. (2017).Lundberg, S.M., & Lee, S. (2017). A Unified Approach to Interpreting Model Predictions. Neural Information Processing Systems.

[74] Sedgwick, P. (2014). Spearman's rank correlation coefficient. *BMJ*. https://doi.org/10.1136/bmj.g7327.

[75] Hollander, M. *et al.* (2015). *Nonparametric Statistical Methods*, Wiley.

[76] Shalabh (2011). Nonparametric Statistical Inference, 5th edn. *Journal of the Royal Statistical Society Series A: Statistics in Society*. https://doi.org/10.1111/j.1467-985X.2010.00681_6.x.